

CLEAR: Colon Segmentation with Liquid Error Avoidance and Refinement for VR Colonography

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ABSTRACT

Colon cancer remains a leading cause of cancer-related deaths, emphasizing the importance of early detection and accurate diagnostics. While colonoscopy is the gold standard, its invasiveness deters frequent, widespread use. Desktop colonography offers a less invasive alternative, and Immersive Colonography improves the efficiency of the analysis. However, colonography requires anatomical reconstruction from CTs, and liquid pockets in CT colonography often obscure the colon, necessitating dual-position scans and doubling radiation exposure. We introduce an automatic segmentation method that integrates liquid regions to eliminate the need for a second CT scan. By requiring only one CT image, this approach enhances patient safety and comfort while maintaining accurate segmentation. The proposed method uses advanced filtering and geometry-based techniques to accurately segment air and liquid pockets, ensuring continuity in colon reconstructions. This innovation promises safer, more efficient VR colonography and improved diagnostic capabilities.

Index Terms: Colonography, Colon Segmentation, Medical Imaging

1 INTRODUCTION

Colon cancer is one of the leading causes of cancer-related deaths worldwide [19]. Early detection results in more effective treatments, underscoring the need for effective and efficient diagnostic tools. Total Colonoscopy (TC) has been the gold standard for detecting colon polyps and potentially diagnosing them as cancer [2, 16]. However, it is often perceived as invasive and uncomfortable by many patients, potentially leading to the underutilization of this critical screening method. In response, CT-colonography (CT-C) has emerged as a less invasive alternative. Unlike colonoscopy, colonography provides a non-invasive, patient-friendly procedure for examining the colon. Furthermore, recent advances allowed researchers to perform colonography in VR, with a demonstrated potential to significantly reduce the time spent on diagnostic data analysis and, consequently, the cost [17, 18].

Segmentation is a crucial step in either desktop or VR-colonography, as it allows clinicians to isolate and reconstruct the colon for detailed analysis. In CT colonography, patients often need to do a dietary preparation and take a contrast dye; usually, the bowels are inflated during the CT scan for better visualization and contrast, significantly increasing patient discomfort. In these cases,

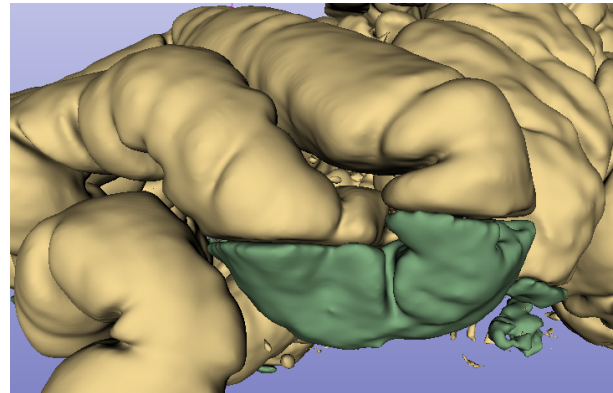


Figure 1: 3D reconstruction of the colon in yellow with the missing segment due to a liquid pocket in green

simple thresholding often results in quality results. Segmentation without bowel inflation is still a challenging problem, although recent advances allow sufficient contrast for segmentation using neural network-based methods [12].

One of the key issues in colonography is minimizing patient exposure to radiation, especially when using computed tomography (CT) scans. Magnetic resonance colonography (MRC) is a promising alternative, however, MRI images typically lack the natural contrast seen in CT images, making it harder to segment the colon accurately.

For CT scans, most of the colon can be segmented using a simple contrast operation. However, many times, there are liquid pockets that obstruct or completely block the colon, as presented in Fig. 1, also, polyps in liquid pockets or at the boundary of air and liquid might not be included in the segmentation, causing false negatives. To avoid this problem, two CT scans are taken in different positions, *ventral decubitus* and *dorsal decubitus*, so that gravity displaces the liquid. Besides the extra time taken, this practice doubles the radiation exposure.

The quality of segmentation in CT colonography is essential because it affects the accuracy with which we can identify polyps. Good segmentation helps us see all the structures in the colon clearly, making it easier to spot polyps and other issues. Common pitfalls of colon segmentation, such as included feces or air bubbles, can be mistaken for polyps, leading to false positives [13].

In this work, we propose an automatic method to include the liquid in the segmentation removing the need for the second CT

scan.

The main contributions of this work are as follows:

Development of a fully automated pipeline for colon segmentation, using image filtering techniques to enhance contrast, including liquid pockets, without requiring additional scans or human interaction.

Integration of air and liquid pocket segmentation into a unified reconstruction, with smoothing techniques to address boundary defects.

2 RELATED WORK

Image segmentation is a critical task in medical imaging, enabling the identification and isolation of anatomical structures for disease detection, treatment planning, and surgical guidance. Various approaches have been developed in colon segmentation to address challenges such as liquid obstructions in virtual colonography.

Geometry-Based Methods. Traditional segmentation techniques rely on geometric properties such as shape and intensity gradients. Thresholding is a simple yet widely used method that segments high-contrast structures by converting images into binary labels. However, it struggles with low-contrast regions, artifacts, and multimodal imaging. More advanced methods, such as edge detection algorithms (e.g., Sobel and Canny filters [14]), improve segmentation by identifying intensity gradients and edge continuity. Region-growing techniques expand from seed points based on intensity similarity, as demonstrated by Franaszek et al. [3], who combined thresholding and fuzzy connectedness to segment air and liquid pockets in CT colonography. Lu et al. [10] further automated region-growing by using geodesic distances to compute the colon's centerline, eliminating the need for manual seed points. Shape detection methods, such as tubularity detection [5, 11], leverage prior knowledge of anatomical structures to improve accuracy.

Neural Network-Based Methods. Neural networks, particularly convolutional neural networks (CNNs), have revolutionized medical image segmentation by learning complex patterns from large datasets. U-Net [7] and its variants (e.g., nnU-Net [8]) are widely used due to their encoder-decoder architecture and skip connections, which capture both fine details and global context. nnU-Net, in particular, automates network configuration based on dataset properties, achieving state-of-the-art results. Large datasets, such as TotalSegmentator [20] and 8K Atlas [9, 15], have been created using semi-automatic segmentation, where radiologists refine AI-generated segmentations to improve quality.

Transformer-Based Methods. Transformers, such as Vision Transformers (ViTs) [6] and Swin Transformers [1], excel at capturing long-range dependencies and global context. However, they often underperform on smaller datasets compared to U-Nets. Hybrid architectures, such as nnFormer [21] and SAM2 [22], combine U-Net and transformer features to overcome these limitations, offering improved performance in medical imaging tasks.

While neural networks and transformers excel in general-purpose medical image segmentation, geometry-based methods remain highly effective for specific challenges, such as liquid obstructions in colon segmentation. Liquid pockets in CT colonography disrupt colon continuity, often requiring a second scan to resolve obstructions. Neural networks struggle in this context due to the lack of large, annotated datasets. Geometry-based methods, like those by Franaszek et al. [3], address these issues by leveraging spatial relationships and intensity patterns, combining thresholding and fuzzy connectedness to segment air and liquid pockets.

Building on these principles, this work develops a fully automated pipeline, eliminating the need for additional scans or manual input.

3 METHOD

We propose a fully automated pipeline for colon segmentation, removing liquid pockets that obstruct or disrupt its continuity. The code to our implementation of the pipeline is available on Github¹.

Our proposed pipeline adopts a fully geometry-based approach, sharing similarities with the method presented by Franaszek et al. [3]. Contrarily to many current approaches, ours does not involve using CNNs. Moreover, it goes one step further by dismissing human interaction and incorporating image filtering techniques to enhance contrast, which leads to a more robust segmentation.

The pipeline illustrated in Fig. 2 begins with the initial segmentation of air-filled regions through thresholding. Unwanted components are filtered out, and flat boundary layers between air and liquid regions are geometrically identified. An image filter is then configured to enhance the contrast of higher-density regions, facilitating the segmentation of liquid pockets using region growth with a defined threshold. Finally, air regions and liquid pockets are integrated, followed by Gaussian smoothing to refine the segmentation and produce a seamless final output. The next sections provide detailed explanations of each step of our pipeline.

3.1 Thresholding Setup

In this initial step, we use a straightforward thresholding operation on the unmodified Ct image with an upper bound of $-800HU$ to reliably identify all air-filled segments, including unwanted structures that must be removed.

To filter out these unwanted elements, while Franaszek et al. [3] builds a tree structure by using a region growing technique which requires a manually placed starting seed, we employ a simple yet effective heuristics that don't require any human interaction. Firstly, a connected components algorithm is applied with components smaller than a predefined size threshold being discarded. Next, the background air layer is identified and removed, ensuring, this way, that relevant structures remain.

Next, we eliminate the layer corresponding to the patient bed. To identify this layer, we begin at the center of the bottom horizontal 2D slice and trace a line upwards. The first component encountered is a consistent patient bed, which can be reliably removed from the segmentation.

Finally, we remove the lungs, as our pipeline is designed for abdominal CT scans where the lungs are typically only partially visible. From our observations, the lungs are truncated at the edges of the image. This characteristic allows us to identify the lung regions by iterating through the top vertical slice, discovering which components are present in that top slice, and excluding them from the final segmentation.

After this first step, the result is that the air parts of the colon are segmented; this can be only one component or many if the liquid pockets break the colon continuity.

3.2 Boundary Layers

To effectively detect and merge the liquid pockets with the air pockets, it is essential to identify the boundary regions between them. Franaszek et al. [3] achieve this by detecting a transitional "sandwich-like" pattern at the boundary of air and liquid pockets. In contrast, our approach identifies flat, downward-facing surfaces within the air pockets and subsequently refines these detections using the Sobel filter for enhanced boundary clarity.

The process begins by calculating a gradient matrix using the Sobel filter to identify surface orientations. Next, we filter the image based on gradient similarity to the target surface normal and group the results into connected components. Finally, we apply a size threshold to remove small, irrelevant surfaces, retaining only the significant components we will call flat labels.

¹GitHub repository: <https://github.com/PedroSFreire/GeometricalColonSeg>

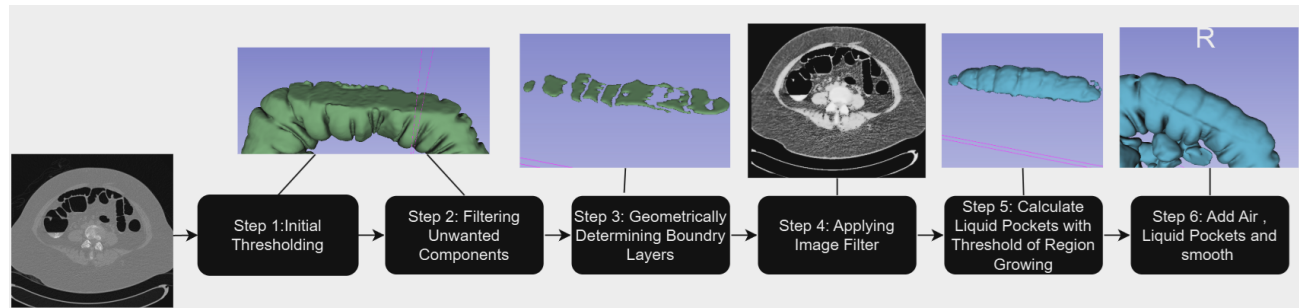


Figure 2: Flowchart of the proposed pipeline, with images for the segmented air and liquid pockets, the detected boundary layers, and the final result.

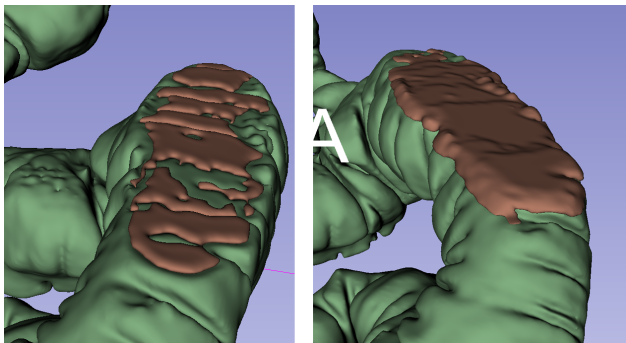


Figure 3: Reconstruction of air pockets of the colon. On the left with the flat label before expansion. On the right with the flat label region growing.

At this stage, the flat labels remain imperfect. After segmenting the liquid pockets, labels intersecting a liquid pocket undergo a region-growing process guided by the Sobel filter matrix, while all other labels are discarded. To address potential gaps in the segmentation, any unselected voxel located between a label voxel and a liquid pocket voxel is added to the label, ensuring continuity and minimizing segmentation holes. The result can be seen in Fig. 3.

3.3 Segmentation of Liquid Pockets

Segmenting liquid pockets presents significant challenges. The contrast liquid often settles at the bottom of the pockets, making it difficult to achieve consistent segmentation (Figs. 4-right and 5-right). Additionally, low contrast between liquid and surrounding dense organs can cause leakage into those regions. Simple thresholding may also inadvertently include bones and other dense tissues, further complicating the segmentation process.

To address this issue, we apply filters to enhance contrast before segmenting the liquid pockets. The process begins with intensity clamping, which reduces noise and isolates relevant tissue types by restricting the intensity values to a meaningful range. Next, histogram equalization enhances contrast within the liquid pockets, making them more distinguishable. Finally, an anisotropic filter is applied to reduce noise while preserving edges, ensuring a clearer and more accurate segmentation. An example of the combined result of the filter can be seen in Fig. 4.

After enhancing the image, we segment the liquid pockets using one of the two available methods. The first one is a simple thresholding operation which now yields significantly improved results due to the enhanced contrast as seen in Figure 5. The second alternative is a region growing technique which allows for accurately delineating

the liquid pockets.

3.4 Segmentation of Pockets Using Thresholding

We observe promising results when applying simple threshold segmentation with a lower bound of $200HU$ to the enhanced images to isolate the liquid pockets. However, several issues persist, many of which can be addressed with further refinement.

The primary challenge is the inclusion of unwanted structures such as bones, liquid pockets in the small bowel, and other soft tissues. To mitigate this, we leverage the previously identified flat labels from the air-filled regions of the colon.

The refinement process begins by running a connected-components algorithm. Components intersecting with the previously calculated flat labels are retained, ensuring only relevant structures are included. Additionally, the detected flat labels are incorporated into the segmentation to assist in defining the air-liquid boundary more accurately.

3.5 Segmentation of Pockets Using Region Growing

Region-growing algorithms are a common approach in medical image segmentation, they work by iterating through each voxel in an initial seed volume and examining its neighboring voxels for potential expansion [4]. The initial seed volumes in our pipeline are derived from the previously calculated flat labels.

The expansion criterion varies based on the desired segmentation precision. The simplest criterion is a straightforward threshold operation. This can be enhanced by incorporating similarity to the expanded voxel, which improves noise resistance. This approach allows the expansion to traverse smooth gradients while halting at edges, resulting in more accurate and robust segmentation.

The main issue identified with the region's growth is the occurrence of expansion leaks. These leaks are managed by halting and removing expanding volumes once they exceed a certain threshold, but this approach can sometimes lead to the removal of valid liquid pockets.

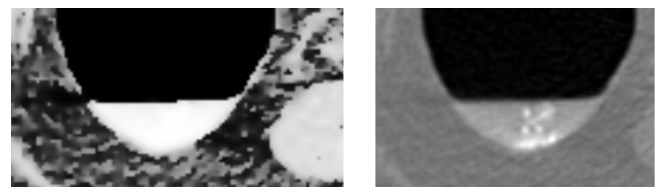


Figure 4: On the left, the contrast-enhanced ct on the right, the original. We can see the settled contrast is mostly gone.

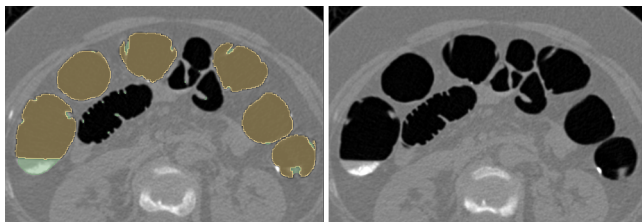


Figure 5: On the left, a segmentation of the colon is shown by a mask overlay highlighting air pockets in yellow and liquid pockets in green. This was achieved using our method. The image on the right is the unprocessed CT scan, where the heterogeneity of the contrast within the liquid pockets is more evident.

Smoothing After we combine the air and liquid pockets, we are left with a defect in the boundary layers. To help fix this and ensure the volumes are connected, we already include the flat labels in the segmentation. This still leaves seams, which we improve by running a Gaussian smoothing filter.

4 EXPERIMENTAL SETUP

The dataset used for evaluation is extracted from the Colon CT dataset². This dataset contains CT scans of the colon, but no ground truth segmentation data is available, which limits the ability to provide quantitative metrics. Consequently, our evaluation is based on a qualitative visual inspection of the segmentation results.

The pipeline was implemented in Python and operates sequentially. The experiments were conducted on a laptop with an AMD Ryzen 7 7840HS processor, an NVIDIA GeForce RTX 4060 GPU, and 16 GB of RAM. While the current implementation does not utilize GPU acceleration, the hardware setup provides sufficient computational resources for testing and evaluation. Each CT scan has a resolution of 512x512x625, and the runtime for processing a single scan is approximately 40 minutes with this setup.

For validation, three CT volumes from the dataset were processed and analyzed. The evaluation focused on the framework's ability to segment air and liquid pockets, handle liquid obstructions, and maintain the continuity of the colon in 3D reconstructions. The results were assessed qualitatively through visual inspection supervised by a radiologist.

5 RESULTS

Fig. 5 shows a 2D slice of the CT scan, where the air pockets in the colon are correctly segmented. The 3D reconstruction highlights the continuity of the segmented colon, with air pockets in yellow and liquid pockets in green. The method successfully integrates liquid obstructions into the segmentation, addressing one of the key challenges in colon segmentation without requiring a second CT scan.

Fig. 6 provides additional visual evidence of the segmentation quality, showing both internal and external views of the colon. The method effectively handles low-contrast regions and complex boundaries, ensuring liquid pockets are accurately segmented and integrated with air-filled regions.

The current implementation runtime of approximately 40 minutes per CT scan is a limitation for clinical applications, although it is a batch process that occurs without human intervention. The most computationally expensive steps are region growing and connected components analysis. However, the algorithm is parallelizable, and GPU optimization is expected to reduce runtime.

Our experiments resulted in the following key observations:

²available at: www.cancerimagingarchive.net/access-data/

Segmentation Accuracy: The method successfully segments air and liquid pockets, even in challenging cases with low contrast or complex boundaries. The integration of liquid obstructions into the overall segmentation is visually accurate. As this work is ongoing, we still have not run specific metrics.

Continuity: The 3D reconstructions demonstrate good continuity of the segmented colon, with minimal artifacts or disconnected components.

Limitations: Some minor artifacts and over-smoothing at air-liquid boundaries were observed, which could be addressed in future refinements.

These results demonstrate the potential of the proposed method in segmenting air and liquid pockets while maintaining the continuity of the colon. However, some limitations remain, which highlight areas for future improvement.

6 DISCUSSION

The proposed method demonstrates strong potential for automated colon segmentation, particularly in its ability to accurately segment air and liquid pockets while maintaining the continuity of the colon in 3D reconstructions. These results highlight the approach's effectiveness in addressing key challenges in colon segmentation, such as handling liquid obstructions and ensuring smooth transitions between segmented regions. While there are areas for further refinement, the current implementation provides a solid foundation for future advancements. Below, we discuss the strengths of the method and opportunities for improvement in three key areas: segmentation quality, computational performance, and algorithm robustness.

Segmentation Quality. The method accurately calculates air and liquid pockets, even in challenging cases with low contrast or complex boundaries. The integration of liquid obstructions into the overall segmentation is a notable strength, as it eliminates the need for additional scans or manual intervention. The 3D reconstructions demonstrate good continuity, with minimal artifacts or disconnected components, critical for clinical applications such as virtual colonoscopy. However, some minor artifacts were observed, such as low-quality seams at air-liquid boundaries and occasional smoothing artifacts. These issues are relatively minor and do not detract significantly from the overall segmentation quality but could be further refined to enhance the method's precision. Future work could incorporate advanced boundary refinement techniques, such as horizontal region expansion methods [3], to improve the delineation of air-liquid interfaces.

Computational Performance. The current implementation operates sequentially in Python, resulting in a runtime of approximately 40 minutes per CT scan. This is a significant limitation for clinical applications, where real-time or near-real-time processing is often required. The most computationally expensive steps, such as region growing and connected components analysis, are particularly well-suited for parallelization. To improve performance, the algorithm should be optimized for GPU acceleration. By leveraging the parallel processing capabilities of modern GPUs, the runtime could be reduced, making the method more practical for clinical use.

Algorithm Robustness. The algorithm's robustness is another key strength, as it performs well across various cases, including those with low contrast or complex anatomical structures. Using contrast enhancement filters and region-growing techniques ensures that air and liquid pockets are accurately segmented in most scenarios. However, there is room for further refinement to make the method even more adaptable to diverse datasets and challenging conditions. For example, while effective, using fixed thresholds and basic intensity-based filters may not generalize optimally to all cases. Incorporating more sophisticated expansion techniques, such as those based on Gaussian or cosine similarity metrics, could

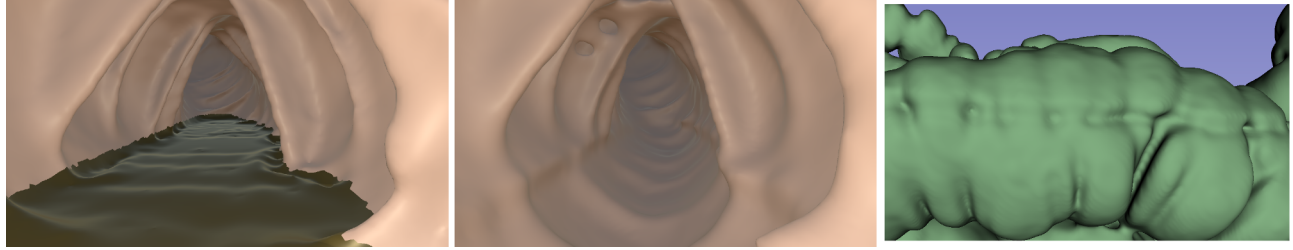


Figure 6: The reconstruction as viewed in VR. Left: The result of a simple air-tissue threshold-based segmentation, which fails to include the liquid pockets. Center: The 3D reconstruction produced by our segmentation algorithm merging the air and liquid pockets. Right: The external view of our reconstruction shows light artifacts at the seams of the liquid pocket.

improve the accuracy of region growth and reduce the occurrence of expansion leaks.

7 CONCLUSION

The proposed method represents a significant advancement in automated colon segmentation, delivering accurate and visually consistent results in segmenting air and liquid pockets. Its ability to effectively handle liquid obstructions and maintain continuity in 3D reconstructions demonstrates the method's potential for addressing key challenges in this domain. While improvements are needed in runtime optimization and artifact reduction, the current implementation establishes a solid foundation for future development. With further refinement and validation, this approach can potentially make VR colonography a reliable tool for clinical applications, enhancing workflows and contributing to better patient outcomes.

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